

Lab 07: Adaptive Maintenance of a Legacy System

In this lab, you will perform adaptive maintenance on an archived, legacy open-source project called Codeface (originally developed by Siemens).

The Challenge

Project Repository: <https://github.com/siemens/codeface>

Original Documentation: <https://siemens.github.io/codeface/>

Codeface is a framework for analyzing the social and technical organization of software development projects. It was archived in 2022 and relies on older versions of Python, R, specific database engines, and outdated dependency trees.

Your goal is to successfully configure, patch, and execute Codeface on your modern local machine so that it can successfully ingest and analyze a sample target repository (such as a small, well-known open-source project like <https://github.com/pallets/flask>).

Success Criteria

The lab is considered complete when you can prove that:

1. The project's core dependencies have been successfully installed and linked.
2. The required backing services (e.g., databases) are running and communicating with the application.
3. You have successfully executed a Codeface analysis run on a target git repository.
4. The system has generated the expected output (e.g., populated database tables or generated analysis files) without fatal crashes.

Deliverables

Your submission must include:

- A detailed, chronological narrative of your debugging and fixing process.
- Provide screenshots of your terminal showing a successful, error-free analysis run on your chosen target repository.
- Provide evidence of the output (e.g., a query result from the populated database, generated sociograms, or the running web frontend).
- AI Declaration